Product Specification & Requirements Document (PRD) — v1.0 · 16 July 2026

TEK Restaurant Intelligence Platform

Client	TEK — Tasty E-Kitchen Ltd, London, UK (tekvers.ai)
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1. Executive Summary

TEK positions itself as "the intelligence layer that powers the future of hospitality." This document specifies the buildout of the core platform that delivers that promise: a multi-tenant SaaS for independent restaurants and hospitality groups, built from scratch, comprising four integrated modules:

1. **First-party customer food-preference database** — TEK-owned capture surfaces that turn every guest interaction into structured preference data.
2. **AI meal recommendation engine** — personalised dish recommendations per guest, per venue, with human-readable reasoning.
3. **CRM with automated personalised marketing and loyalty programmes** — segmentation, automated campaigns (email/SMS), and a configurable loyalty engine.
4. **AI predictive analytics** — churn risk, demand forecasting, and menu-performance intelligence surfaced in a venue dashboard.

The platform is delivered in three fixed-price phases (plus an optional fourth for POS integrations), each independently valuable and demonstrable. Two pricing tracks are quoted: **Lean** and **Studio** (Section 12).

Out of scope for this engagement: VR food visualisation/commerce, native mobile apps, and hardware/smart-kitchen infrastructure. The architecture leaves clean seams for all three.

2. Background, Problem & Goals

2.1 Problem

Independent restaurants lose the retention game: the guest relationship is owned by aggregators and generic booking tools. Venues rarely know *who* their guests are, *what* they like, or *when* they are about to lapse. TEK's own market framing cites customer-retention struggles affecting 86% of hospitality businesses.

2.2 Product goals

- **G1** — Give every TEK venue a first-party, GDPR-compliant guest database it owns, populated automatically from TEK capture surfaces.
- **G2** — Convert that data into measurable revenue actions: recommendations that lift average order value, campaigns that lift visit frequency, loyalty that lifts retention.
- **G3** — Give operators forward-looking intelligence (churn, demand, menu performance) rather than backward-looking reports.
- **G4** — Support TEK's commercial model (Starter Intelligence at £249/venue/month; Enterprise custom) with per-venue multi-tenancy, seat management, and usage isolation from day one.

2.3 Success criteria (platform-level)

Metric	Target (first 6 months post-launch)
Guest profiles captured per active venue	≥ 500
Guests with ≥ 3 preference signals	≥ 60% of profiles
Recommendation click-through (QR menu)	≥ 15%
Campaign-attributed repeat visits	measurable per campaign, ≥ 8% redemption
Churn-model precision @ top decile	≥ 2× baseline repeat rate
Dashboard weekly active usage	≥ 70% of paying venues

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3. Scope

3.1 In scope

- Multi-tenant SaaS core: organisations → venues → seats (role-based access), venue onboarding wizard.
- Guest-facing capture surfaces: QR web menu, web ordering page, signup/consent forms, post-visit feedback prompts, loyalty join flow.
- Customer food-preference database (Module 1) with a unified guest-event spine.
- AI meal recommendation engine (Module 2).
- CRM, campaign automation (email + SMS), and loyalty engine (Module 3).
- Predictive analytics: churn scoring, demand forecasting, menu-performance analytics (Module 4).
- Venue-facing business intelligence dashboard.
- GDPR/UK-DPA compliance tooling: consent records, export, right-to-erasure.
- TEK internal admin panel (tenant management, plan limits, platform health).

3.2 Out of scope (this engagement)

- VR/3D food visualisation and VR commerce.
- Native iOS/Android apps (all guest surfaces are mobile-first web).
- Payments/ordering fulfilment beyond capture (no kitchen display, no delivery logistics).
- POS integrations in Phases 1–3 (delivered as optional Phase 4).
- Multi-language localisation (English-first; i18n-ready string handling).

3.3 Assumptions

- TEK supplies brand assets, domain(s), and legal copy (privacy policy, terms) — templates provided by OEFR.

- TEK operates the commercial relationships with venues; OEFR delivers the platform.
- Third-party service costs (hosting, email/SMS delivery, LLM API usage) are billed directly to TEK's accounts; estimates in Section 11.
- Pilot cohort of 3–10 venues available for Phase 2–3 validation.

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4. Users & Personas

Persona	Description	Primary needs
Venue Owner/GM	Runs 1–3 sites, time-poor	Retention, revenue lift, zero-admin tooling
Marketing Manager	Group/franchise level	Segments, campaigns, loyalty configuration, attribution
Front-of-house staff	Service staff	Guest preference card at a glance (allergies, favourites, VIP)
Guest (diner)	End customer	Fast menu/ordering, relevant recommendations, rewards worth joining, control over their data
TEK Admin	TEK's own team	Tenant provisioning, plan enforcement, platform health, model performance

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5. Module 1 — First-Party Customer Food-Preference Database

The foundation. Every other module reads from this.

5.1 Concept: the guest-event spine

All guest activity is written as immutable events against a guest profile: `menu_view`, `dish_view`, `order_placed`, `order_item`, `feedback_given`, `campaign_open`, `campaign_click`, `reward_redeemed`, `visit_checkin`, `preference_declared`. Preferences are **derived** (from behaviour) and **declared** (explicitly stated), kept distinct and both queryable.

5.2 Capture surfaces (v1, TEK-owned)

Surface	Captures
QR web menu (per venue/table)	Menu/dish views, dwell, session device link
Web ordering page	Orders, items, modifiers, spend, time-of-day
Signup & loyalty join flow	Identity (email/phone), declared preferences, dietary needs, marketing consent
Post-visit feedback prompt (email/SMS link)	Dish ratings, visit rating, free-text comment
Staff quick-note (dashboard)	FOH-entered notes: allergies, occasions, VIP flags

5.3 Guest profile

- Identity: email and/or mobile (E.164), name optional; anonymous sessions merge into a profile on identification (deterministic matching only).
- Declared: dietary requirements (vegan, halal, gluten-free, allergens list), cuisine likes/dislikes, spice tolerance, favourite dishes, occasions (birthday, anniversary).

- Derived: taste vector (per taxonomy tags), price-band affinity, visit cadence, channel preference, RFM scores (recency/frequency/monetary).
- Consent object: per-channel marketing consent with timestamped audit trail; profiling opt-out flag honoured by Modules 2–4.

5.4 Menu & taxonomy

- Venue menu manager: categories, dishes, prices, photos, availability windows.
- TEK food taxonomy: every dish tagged (cuisine, ingredients, allergens, spice level, dietary flags, preparation) — tags proposed automatically by LLM from the dish name/description, confirmed by the venue in onboarding.

5.5 Acceptance criteria (Module 1)

- A guest can scan a QR code, browse the venue menu, and their session events persist and merge into their profile after signup.
- A venue can view, search, filter, and export its guest list; guests are strictly invisible across tenants.
- A guest can request their data (export) and erasure; erasure completes across all modules within 30 days and is logged.
- Preference records distinguish declared vs derived, each with provenance and timestamps.

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6. Module 2 — AI Meal Recommendation Engine

6.1 Behaviour

- **Known guest, on QR menu/ordering page:** “For you” rail of 3–6 dishes ranked by taste-vector match, order history, dietary constraints (hard filters), availability, and margin weighting (venue-configurable).
- **Unknown guest:** venue-level popularity + time-of-day priors (cold start), improving within the session from dish views.
- **Explanations:** each recommendation carries a short human-readable reason (“Because you loved the Jollof Special”), LLM-generated from structured signals — never invented facts.
- **Staff view:** FOH preference card shows the same top picks for table-side upsell.

6.2 Engine design

- Stage 1 (retrieval): hard constraint filtering — allergens, dietary flags, availability. **Allergen safety is absolute: a dish containing a declared allergen is never recommended, and the filter is enforced in code, not by the model.**
- Stage 2 (ranking): weighted hybrid — item-based collaborative filtering on order events + content similarity on taxonomy tags + recency/novelty terms. Deterministic, unit-testable scoring.
- Stage 3 (presentation): LLM formats explanation copy from the scoring evidence.
- Feedback loop: impressions and clicks logged as events; weekly offline evaluation (CTR, add-to-order rate) per venue.

6.3 Acceptance criteria (Module 2)

- Recommendations respond < 300 ms p95 (excluding explanation copy, which streams after).
- A guest with a declared allergen never sees a dish containing it (verified by automated test across full menu permutations).
- Venue can toggle margin weighting and see recommendation CTR in the dashboard.
- Cold-start venues get sensible popularity-based rails from day one.

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7. Module 3 — CRM, Automated Marketing & Loyalty

7.1 CRM & segmentation

- Segment builder over profile + event data: attributes (dietary, preferences, RFM band, churn-risk band from Module 4), behaviours ("ordered ≥ 2× in 30 days", "viewed menu but didn't order"), and consent state (always enforced).
- Saved segments update dynamically; counts preview before sending.

7.2 Campaign automation

- Channels: **email + SMS** in v1 (Resend/Postmark for email, Twilio for SMS — final vendor choice at Phase 3 kickoff).
- Campaign types: one-off blasts, and **automations** triggered by events/conditions: welcome series, lapsed-guest win-back (churn-risk triggered), birthday/occasion, post-visit feedback ask, reward-earned notification.
- Personalisation: merge fields + AI-drafted copy variants per segment (venue approves before send; nothing auto-sends unreviewed copy in v1).
- Attribution: per-campaign UTM/short-link + redemption codes; dashboard reports opens, clicks, visits/orders attributed, revenue attributed.
- Compliance: per-channel consent enforced at send time; one-click unsubscribe; suppression lists; PECR-compliant sender identities.

7.3 Loyalty engine

- Configurable per venue (or shared across a group — TEK's multi-venue selling point): points per £ spent, visit stamps, or tiered (Bronze/Silver/Gold).
- Rewards: free item, tier perks, occasion bonuses. **No percentage-discount mechanics required for v1** — reward catalogue is value-add oriented.
- Guest experience: join in one step from any capture surface; balance visible on menu page; redemption via short code shown to staff (staff confirm in dashboard).
- AI assist: reward suggestions per segment based on preference data ("Top reward for your Friday regulars: free dessert — 68% of them order dessert").

7.4 Acceptance criteria (Module 3)

- A venue can create a segment, launch an automated win-back campaign, and see attributed redemptions end-to-end.
- Consent revocation stops all sends to that guest within minutes.
- A guest can join loyalty, earn on an order, see balance, and redeem in-venue with staff confirmation.
- Every send is logged (who, what, when, channel, consent basis).

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8. Module 4 — AI Predictive Analytics

8.1 Capabilities

Capability	Description	Method (v1)
Churn risk	Per-guest lapse probability, banded (healthy / cooling / at-risk / lost)	Gradient-boosted classifier on recency/frequency/monetary + engagement features; retrained weekly per tenant cohort
Demand forecast	Expected orders/covers by day and daypart, 14-day horizon	Time-series model (seasonality + trend + holiday calendar); per venue
Menu performance	Dish-level views → orders conversion, attach rate, margin contribution, rising/falling movers	Deterministic analytics over the event spine
Preference trends	Shifting taste-tag demand per venue (“vegan mains +22% this quarter”)	Aggregated tag analytics with significance thresholds

8.2 Dashboard

- Venue home: today’s expected demand, at-risk guest count with one-click “send win-back”, top movers on the menu, campaign performance summary.
- Weekly AI digest: plain-English summary of what changed and the single highest-impact suggested action (LLM-written from computed metrics only — every number in the digest traces to a stored metric).
- All predictions display confidence/coverage honestly; models degrade gracefully to heuristics when a venue has thin data, and the UI says so.

8.3 Acceptance criteria (Module 4)

- Churn bands populate for any venue with ≥ 90 days of data; backtest report generated per model version.
- Forecast accuracy tracked and displayed (MAPE vs actuals) — no unmeasured claims.
- Weekly digest delivered to venue owner by email; every figure reproducible from the dashboard.

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9. Architecture & Technical Design

9.1 Approach: modular monolith (approved)

One Next.js (App Router) application serving venue dashboard, guest capture surfaces, and API routes; PostgreSQL as the single source of truth with **row-level security keyed on tenant ID**; a Python worker service for ML jobs; LLM APIs for language tasks. Chosen over microservices (2–3× cost, unneeded at this scale) and third-party CRM assembly (per-contact fees destroy the £249/venue margin; TEK must own the intelligence layer).

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Guests (QR menu / ordering / signup)     Venue staff (dashboard)      TEK admin
         │                                        │                         │
         ▼                                        ▼                         ▼
┌─────────────────────────────────────────────────────────────────────────────┐
│  Next.js app (Vercel) — guest surfaces · dashboard · admin · API routes     │
└──────────────┬───────────────────────────────────────────────┬──────────────┘
               │                                               │
               ▼                                               ▼
     PostgreSQL (managed, RLS multi-tenant)          Job queue (Postgres-backed)
     profiles · events · menus · segments                      │
     campaigns · loyalty · predictions                          ▼
               ▲                                     Python ML worker (VM/container)
               │                                     churn · forecast · taxonomy
```

```
        └── LLM API (OpenAI) ── explanations · copy drafts · digests
   Email (Resend) · SMS (Twilio) · Object storage (dish photos)
```

9.2 Stack

Layer	Choice	Rationale
Web/app/API	Next.js 15+, TypeScript, App Router	Matches TEK's advertised stack; one deploy target
Database	Managed PostgreSQL with RLS	Tenant isolation enforced at the database, not just app code
ML worker	Python 3.12 (scikit-learn, statsmodels/prophet-class), containerised	Right tool for churn/forecast; isolated from web path
Queue/jobs	Postgres-backed queue (e.g. pg-boss)	No extra infrastructure at this scale
LLM	OpenAI API (per TEK stack); provider-abstracted client	Explanations, copy drafts, taxonomy tagging, digests
Email / SMS	Resend or Postmark / Twilio	Deliverability + UK SMS compliance
Hosting	Vercel (app) + one small container host (worker) + managed Postgres	Minimal ops surface
Auth	Email/password + magic link; seat-based RBAC (Owner/Manager/Staff/TEK-Admin)	No social-login dependency

9.3 Data model (core tables, abbreviated)

`organizations`, `venues`, `seats/users`, `guests`, `guest_identities`, `consents`, `events` (partitioned, append-only), `menus`, `dishes`, `dish_tags`, `preferences` (declared/derived, provenance), `segments`, `campaigns`, `sends`, `loyalty_programs`, `loyalty_accounts`, `loyalty_transactions`, `rewards`, `redemptions`, `predictions` (model, version, score, band, computed_at), `model_runs` (metrics, backtests).

9.4 Security & non-functional requirements

- **Tenant isolation:** Postgres RLS on every tenant-scoped table; cross-tenant access is a test-suite failure class of its own.
- **PII:** encrypted at rest (managed-DB level) and in transit; PII columns access-audited; secrets in environment config, never in code.
- **GDPR/UK DPA:** consent audit trail, subject access export (JSON/CSV), erasure workflow (hard-delete PII, pseudonymise events), data-processing register documentation handed to TEK.
- **Performance:** guest surfaces LCP < 2.5 s on 4G; recommendation API < 300 ms p95; dashboard queries < 1 s p95.
- **Availability:** target 99.9% on managed platforms; graceful degradation — if ML worker or LLM is down, menus/ordering/loyalty continue (recommendations fall back to popularity, digests skip a week).
- **Observability:** structured logs, error tracking (Sentry-class), per-tenant usage metering (supports TEK's plan enforcement).
- **Testing:** unit + integration suites per module; the allergen-filter and consent-enforcement paths get exhaustive automated tests; each phase ends with a UAT script executed with TEK.

10. Delivery Plan & Timeline

Phase	Contents	Duration	Demo/exit criteria
1 — Data Foundation	Multi-tenant core, auth/RBAC, venue onboarding, menu manager + AI taxonomy tagging, QR menu + ordering + signup surfaces, guest profiles + event spine, GDPR tooling, TEK admin panel	5 weeks	A pilot venue onboards itself, guests browse/order/sign up, profiles populate, erasure works
2 — Intelligence	Recommendation engine (all 3 stages), analytics dashboard, churn model, demand forecast, menu performance, weekly digest	5 weeks	Live recommendations on the pilot venue's menu; churn bands + forecasts on real captured data; backtest reports
3 — Activation	Segment builder, campaign automation (email+SMS), attribution, loyalty engine + guest surfaces + staff redemption, AI copy/reward assist	5 weeks	End-to-end: segment → automated campaign → attributed redemption; loyalty earn/redeem live
4 — POS Integrations *(optional)*	Square, Toast or Lightspeed connectors (2 in scope): historical import + ongoing order sync into the event spine	4 weeks	Pilot venue's POS orders flow into profiles and retrain models

Total core programme (Phases 1–3): **~15 weeks** from kickoff. Each phase gates on written acceptance before the next begins.

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11. Running Costs (TEK's accounts, estimates)

Item	Est. monthly (pilot scale, ≤ 25 venues)
Hosting (Vercel Pro + worker host + managed Postgres)	£90–£180
Email (per-send)	£15–£60
SMS (Twilio UK, usage-based)	£0.04/msg — campaign dependent
LLM API	£30–£120 (explanations cached; copy drafts on demand)
Error tracking / monitoring	£0–£25

Per-venue infrastructure cost at pilot scale: **≈ £6–£15/month**, comfortably inside the £249/venue price point.

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12. Commercial Proposal — Fixed Price per Phase

Two tracks. Same scope and acceptance criteria per phase; tracks differ in delivery wrap (documentation depth, UAT support, warranty, response SLAs).

Track A — Lean

Solo senior delivery, AI-accelerated. 30-day post-phase defect warranty, async support (48 h response), essential documentation (admin guide + API reference).

Phase	Price
1 — Data Foundation	£12,000
2 — Intelligence	£14,000
3 — Activation	£12,000
Core total (1–3)	**£38,000**
4 — POS Integrations (optional)	£9,000

Track B — Studio

Everything in Lean, plus: dedicated staging environment with seeded demo tenant for TEK sales use, extended UAT (2 rounds per phase + on-call launch support), 90-day warranty per phase, full documentation pack (architecture dossier, runbooks, onboarding playbooks, training videos), priority SLA (same-business-day), and quarterly model-performance review through the engagement.

Phase	Price
1 — Data Foundation	£26,000
2 — Intelligence	£32,000
3 — Activation	£26,000
Core total (1–3)	**£84,000**
4 — POS Integrations (optional)	£18,000

Ongoing operations (optional, post-launch, either track)

Plan	Scope	Monthly
Ops Essential	Monitoring, patching, model retraining oversight, 8 h/month improvements	£1,500
Ops Growth	Essential + 24 h/month feature development, campaign/model tuning, monthly performance report	£3,500

Payment terms

- Per phase: **40% to commence, 60% on written acceptance** of that phase's exit criteria.
- Prices exclude VAT (if applicable) and third-party running costs (Section 11).
- Quotation valid 30 days from the date of this document.
- Scope changes handled by written change order against a day rate agreed at signature.

13. Risks & Mitigations

Risk	Mitigation
Thin data at pilot venues weakens models	Heuristic fallbacks shipped first; models activate at data thresholds; honest confidence display
Email/SMS deliverability	Dedicated sending domains, warm-up plan, suppression hygiene from day one
Allergen/dietary errors	Hard-coded constraint filtering (never model-decided), exhaustive automated tests, venue confirmation step on taxonomy
GDPR exposure	Consent-first design, erasure workflow tested in UAT, processing register delivered
Scope drift from TEK's broader vision (VR, hardware)	Fixed per-phase scope with written acceptance gates; change-order process
LLM cost/latency	Explanations cached per (guest-segment × dish); language tasks off the critical path

14. Acceptance & Next Steps

1. TEK reviews this document and selects track (A or B) and optional Phase 4 / Ops plan.
2. Statement of Work + first-phase invoice issued; kickoff within 5 business days of deposit.
3. Weekly written progress reports; phase demos on a shared staging environment.

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